Detection of Criminal Activity using Deep Learning

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2024 International Conference on Science Technology Engineering and Management (ICSTEM) | 979-8-3503-7691-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSTEM61137.2024.10561223

Abstract- The surge in criminal activities presents a formidable challenge for global law enforcement agencies. Traditional crime detection methods often struggle to address the intricate and varied nature of criminal behaviors. Many businesses deploy CCTV systems for continuous observation of people and their activities. The developed system exhibits significant potential for applications in public safety, surveillance, and law enforcement, serving as an intelligent and automated tool for early detection and prevention of criminal activities. Continuous human monitoring of data is nearly impossible due to the workforce and attention required. Automation becomes essential. The project proposed utilizes the deep learning technology, a CNN model is designed to identify harmful behaviour in the videos. To expedite the identification of abnormal events, it is crucial to automate this process, indicating which frame and segment contain the unusual activity. This involves dividing the video into frames and scrutinizing the people and activities in each processed frame. When it detects criminal activity, it can trigger alerts or notifications to the personnel, allowing them to respond promptly.

Keywords—CCTV, public safety, prevention.

I. INTRODUCTION

The recognition of criminal activities is pivotal in person identification, with visual information being a primary source for such determinations [1]. Surveillance videos, whether presented in real-time or for future playback, provide crucial visual data. The recent increase in automation has significantly impacted video analytics, facilitating applications like motion detection, human activity prediction, face identification, abnormal activity prediction, vehicle counting, and crowd counting [2]. In this domain, person identification relies on face recognition and gait recognition [3]. Face recognition, particularly, proves versatile for automated identification in surveillance videos, predicting head orientation and aiding in behavior prediction. When motion recognition and face recognition are combined, it helps with identity verification, presence/absence detection at particular times and places, and person verification [5]. A system that can identify and recognize suspicious behavior also benefits from human interactions, such as subtle touch, head motion detection, and hand gesture recognition. This project presents a face recognition methodology for identifying questionable human behavior. Intelligent algorithms are used in the security and research domains to monitor live videos through the use of video processing. The design of a real-time system considers factors like computational and time complexities. An ideal system would have an algorithm with low hardware resource consumption, low time complexity, and high-performance outcomes [6]. This would make it useful for time-sensitive applications like patient monitoring, bank robbery detection, and identifying and reporting criminal activity at places like train stations. Putting in place a system like an automatic criminal activity detection system helps to minimize these kinds of incidents while also aiding in the identification of criminal activities [8].

Utilizing pre-trained deep learning models is an additional method for identifying firearms in surveillance footage. These models aim to help users experiment with existing frameworks for better results without explicit design, or to learn about algorithms. Five layers make up a typical deep learning neural network: fully connected, Max-Pooling, input and output layers with convolution, and so on. Pre-trained deep learning models are popular because they can be used to overcome limitations in time, memory, and CPU and processor resources. Compared to machine learning, which makes use of explicit design, these pre-trained models yield superior results. When watching large amounts of recordings or keeping multiple footages, human guards may get tired or nod off, missing chances to find unusual criminal intention scenarios that could be captured in multiple footages. Pre-trained deep learning models could be used to solve this problem by recognizing potential threats in public spaces without requiring human interaction.

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II. LITERATURE SURVEY

Crucially, the model needs to discern suspicious human poses to effectively identify questionable human behaviour. Human pose estimation, a prominent focus in computer vision research for more than 15 years, involves recognizing various parts of the human body and tracking their movements. This technology finds applications in AR/VR, gesture recognition, game consoles, and various other devices. Initially, inexpensive depth sensors, also known as motion sensors, were employed for detecting human movement, particularly in game consoles. However, these sensors are limited to indoor use, and determining human activity from depth photos is challenging due to their low resolution and noisy depth data.

Numerous models have been made available recently. A real-time automated surveillance system with tracking capabilities for multiple non-intersecting cameras was proposed by Imed Bouchrika [1]. The primary goals of this work are biometrics and gait-based forensic analysis. Additionally, the work is utilized to track people who are walking using security cameras. The primary use for this is for tracking individuals, despite the recent major issue with personal secrets emerging. In addition, forensic systems and biometric surveillance cameras are used in this work. A new model for human detection using a different location camera is put forth. Thus, we can conclude that the primary means of suspecting any activity is the identification of individuals through the use of cameras based on data from various points of view.

Grega [2] released an algorithm that can identify weapons and knives in CCTV footage and alert the operator or security guard. For knife detection, the algorithm's sensitivity is 81.18% and specificity is 94.93%, with an emphasis on minimizing fake warnings and providing an instantaneous application. In addition, the specificity of the fire alert system is 96.6%, and the movie's various items have a sensitivity of 35.98%.

The author [3] details a method for automatically identifying handguns in surveillance footage. This approach involves recognizing weapons in films, classifying objects as guns or not, and predicting when a crime will occur or not, the system aims to reduce the number of crimes that occur. Models based on RCNN and FRCNN perform better than a sliding window proposal method when compared to the system. Even in low-quality videos, the algorithm can accurately predict when crimes will occur, producing good results

In this work [4], the most advanced weapon detection dataset was used to train the YOLO V3 model. The concept presented by the authors equips a machine with the capability to identify deadly weapons and notify a human administrator when a revolver is observed in the vicinity. The YOLO V3 model, after being trained, demonstrates superior performance compared to the YOLO V2 model in the terms of performance and requires less processing power, according to the experimental data. Improving current surveillance capabilities is urgently needed, and better resources must be made available to allow for the monitoring of human operators' effectiveness.

The evaluation of system outcomes involves extracting characteristics from four distinct categories: object form, material composition, time-frequency signal analysis, and transient pulse response. Following this, two classification strategies are applied in order to distinguish between items that pose a threat and those that do not. Over 90% of features are successfully classified by the framework for combining features and classifying within the new system [5].

Despite recent progress in image-based machine learning, identifying an assailant wielding a knife remains a challenge. The authors [6] propose three methods for automated crime detection, employing diverse datasets containing images of knives to address this concern. The goal is to minimize both false positives and false negatives, concurrently mitigating potential assault intentions. Initially, a sparse and pruned neural network classification model, based on MobileNet, is employed, which can accurately (95%) alert with 2.2 MB of memory to alert a witness to the possibility of a criminal wielding a knife. Secondly, they train a mask RCNN detection algorithm to discern between the hand and the object specifically a knife in a single picture. Assign a degree of probability to their relative locations, enabling the inference of point threat and bounding box localization. Lastly, to reduce misunderstandings about the attacker's goals and refine the threat features, anatomical way-points are assigned by a Pose Net-based model [7].

CNNs' broad object recognition capability and robustness to errors have demonstrated great potential for various applications such as internet of things-based photovoltaic monitoring, face identification, crime detection, self-driving cars, and COVID-19 detection [8].

The scales or factors that are utilized to generate feature maps are referred to as "backbones" in the object detection domain. Because they produce the features needed for object detection, these crucial elements are a crucial part of the feature extractor. [9]. For this, a variety of backbones can be employed, each with advantages and disadvantages of their own. Among many, deep learning convolutional neural network (CNN) backbones in object detection techniques are the visual geometry group (VGG) and residual neural network (ResNet). There are several trade-offs between each of these designs in terms of speed, efficiency, and accuracy. VGG, ResNet, and GoogleNet will be used with the suggested model. The algorithms that

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we suggest can be helpful in notifying the human operator when a gun, or any weapon is been observed. The system aims to reduce the number of crimes by (IJACSA), recognizing firearms in movies, categorizing products as firearms or not, and determining if a crime has been committed or not. Comparing the framework's functioning with a sliding window protocol method reveals that model utilizing RCNN and FRCNN, demonstrate strong performance.

The method [10] used can accurately predict instances of crime even in low-quality videos, producing favourable outcome. Unlike traditional optical flow methods that analyse edge features in just two successive frames, a new approach involves tracklets evolving over extended motion projections. Spatio-temporal cuboid sequences collect statistical data on these tracklets as they move through them [11]. There are some new motivations behind criminal behaviour. First offenders frequently identify different locations prior to crimes that are recorded by cameras in various locations. It is not stored for extended periods of time unless it contains unusual behaviour [12]. Third, the primary reason is nearly always an ambassador who exhibits odd behaviour. It is recommended that an abnormal behaviour database be set up for smart cameras, and that surveillance videos be selected and stored based on the information provided, thereby conserving storage space.

III. System Architecture

The rising incidences of criminal activity, terrorism, and fraud highlights the critical need for strong systems that can identify and stop the criminal activity [13]. An essential part of contemporary security plans, criminal activity detection makes use of sophisticated techniques and modern technology to spot abnormal activity, foresee possible threats, and take proactive action. The suggested system intends to utilize footage captured by CCTV cameras to monitor activities.

- **Video Input:** The system begins by taking in real-world video input from cameras or other sources. This could include footage from surveillance cameras, drones, or any device capable of capturing video.
- **Pre-processing:** The preprocessing module prepares the raw video data for analysis by the CNN model. This may involve tasks such as resizing the frames to a standard size, converting color spaces, normalizing pixel values, and removing noise or artifacts from the video.
- **Feature Extraction using CNN:** The CNN model module is responsible for analyzing the extracted frames to detect criminal activity. It consists of multiple layers of convolutional, pooling, and fully connected layers that learn to extract meaningful features.

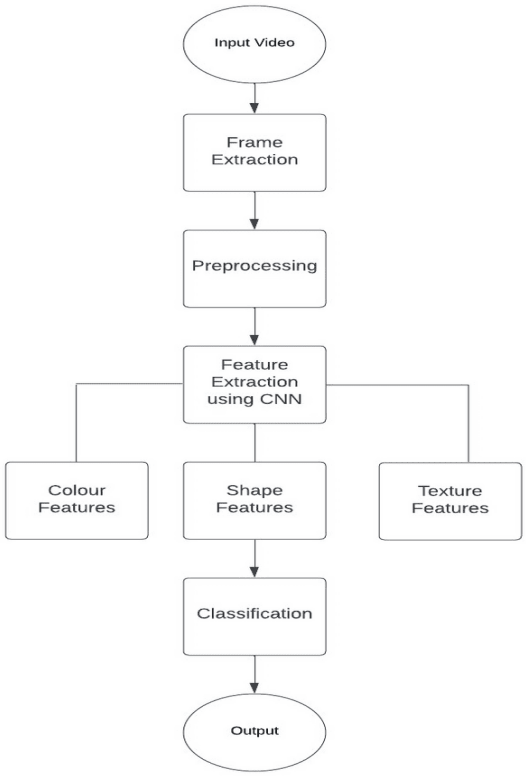

Fig 1: Architecture diagram

- **Classification:** Based on the estimated probabilities, the classification module makes decisions about the classification of each input frame. A threshold should be applied to the probabilities to determine whether a frame is classified as normal or criminal activity
- **Output:** The output module presents the results of the criminal activity detection process to the user. This will include alarm trigger when criminal activity is detected.

III. Existing System

The human operator of the current CCTV surveillance system oversees the system, and the automated system struggles to make decisions and respond appropriately in unusual circumstances.

In dynamic circumstances, monitoring typically focuses on identifying, detecting, and locating a specific thing or object from the given image as well as comprehending the behavior of the object. Our primary goal is to create an intelligent monitoring model system that can replace the current passive CCTV surveillance system, which is becoming more and more inefficient as more cameras are installed and more people need to monitor the system.

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IV. PROPOSED SYSTEM

In this manuscript, we introduce a framework aimed at detecting and tracking criminal activities using surveillance cameras within organizational settings. Typically, surveillance cameras are deployed within various buildings or workplaces to monitor incidents. Consequently, these cameras continuously generate video footage to record the captured data. We utilize this recorded video information and process the data to create discrete individual frames. Subsequently, by processing these individual frames, single images can be generated.

To detect criminal activities, crime inspection data are trained and stored in a database. A crime database serves as a repository for storing classified crime images. Images extracted from the frames captured by the camera are then compared against this trained dataset. Real-time crime images are collected for classification, which involves employing deep learning techniques such as CNN[10] or any other suitable model.

Human criminal activity detection, a field at the intersection of artificial intelligence, machine learning, and video analytics, has emerged as a groundbreaking approach to proactively identify, analyze, and prevent illicit activities. This multifaceted discipline leverages sophisticated algorithms and cutting-edge technologies to scrutinize behavioral patterns, distinguish anomalies, and predict potential threats. As our societies become increasingly complex, the need for advanced tools to assist law enforcement in maintaining public safety has never been more pressing. Human criminal activity detection stands as a beacon of innovation, offering the promise of enhanced security measures and a more vigilant, preemptive approach to combating criminal behavior. The process begins with the acquisition of video data from surveillance cameras. These cameras capturing real-time footage of public spaces, high-risk areas, or locations prone to criminal activities. The collected video data becomes the primary source material for the subsequent stages of analysis. The project is built on the model through algorithms. These algorithms are trained on extensive datasets, encompassing a diverse range of scenarios, behaviors, and activities. During the training phase, the model learns to discern patterns associated with routine, nonthreatening behaviors as well as those indicative of potentially criminal or suspicious activities. The trained model relies on a combination of computer vision techniques and machine learning algorithms to extract meaningful features from the video data. This includes 9 identifying human movements, object interactions, and environmental changes. The system's ability to distinguish between normal and abnormal activities is crucial for accurate detection. As the video data streams in real-time, the system continuously processes and analyzes the incoming frames. The model compares the observed activities against the learned patterns and triggers an alert when it identifies behavior that deviates from the norm.

Convolutional Neural Networks have demonstrated remarkable efficacy in object recognition and image processing over the past few years. CNNs are a type of neural network designed with image recognition in mind. These networks consist of multiple layers, each with a distinct purpose. To teach the network the traits that distinguish each object and associate it with a class, CNN is trained on a sizable dataset of images that include the objects that need to be identified. The network can be trained to identify the necessary objects in fresh images. Figure 2 illustrates the application of a CNN model for weapon detection.

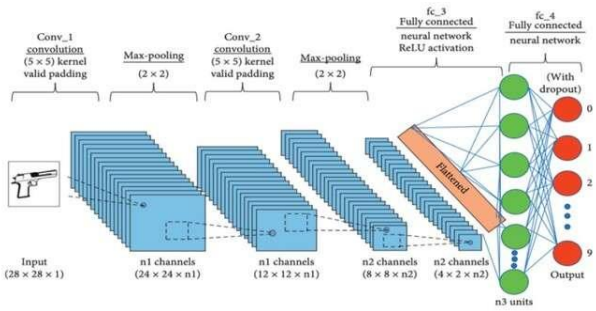

Fig 2: Convolutional Neural Network (CNN)[10].

APPLICATIONS

The rise in criminal activity has made the detection of suspicious activity an important field of research. We looked at the existing methods and provided a different method to identify criminal activities.

- **Security and Surveillance:** Criminal activity detection is widely used in security and surveillance systems to identify unusual behaviours in crowded places, airports, public transportation, and critical infrastructure areas. It helps in preventing crimes, terrorism, and unauthorized access.
- **Retail Security:** To detect employee theft, shoplifting, and other fraudulent activities within their stores, retailers use criminal activity detection. It contributes to minimizing losses and improves general store security.
- **Border Security:** Criminal activity detection is a tool which can be used by border control organizations to keep an eye out for smuggling, illegal crossings, and other suspicious activities. It helps to keep the country safe and prevents illegal border crossings.

V. RESULTS

The project is about classifying the uploaded videos based on the activity occurring in the file. If the activity appears to be any criminal activity, an alert is generated notifying the user that a criminal activity is taking place.

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This system can potentially help detect criminal activity at an early stage, allowing for quicker intervention. By monitoring uploaded videos, it could deter criminal activity. A video upload form enables users to select video files from their devices and submit them to the system for processing. Once analysis is complete, displays the identified activity in clear text. If the system detects a high probability of criminal activity it will trigger an alert indicating an abnormal activity.

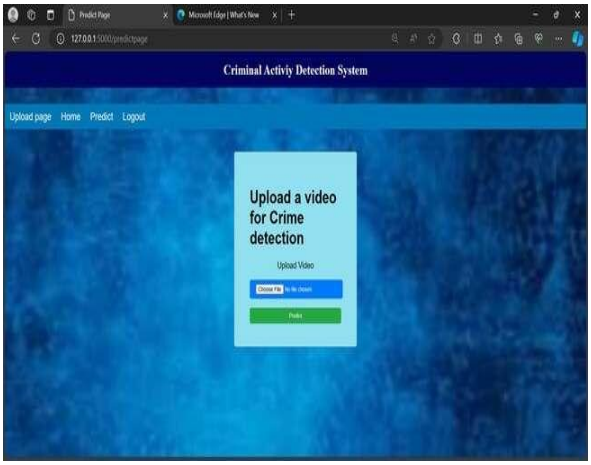

Fig 3: Home Page

The web page consists a login page, where an individual can register. After registering a predict page is directed where the user can upload a video. A predict button will predict the activity.

Fig 4: Result Page

The result page displays the type of activity, and an alert sound is generated if the activity occurred is a criminal activity.

The user can go back to the home page or predict few other videos by clicking “Predict Another Video”. The user credentials are stored in the sql database.

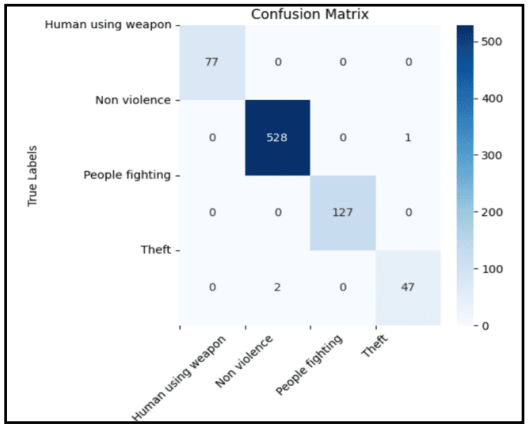

Fig 5: Confusion Matrix

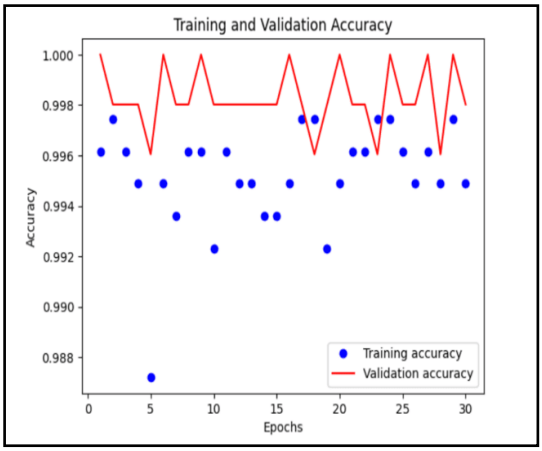

Fig 6: Training and Testing Accuracy

The accuracy plot that was created above gives a visual depiction of the training and validation accuracies during the neural network model training process. The blue dots represent the training accuracy at each epoch and the red line represents the validation accuracy at each epoch.

VI. Conclusion

In today's world, most people recognize the significance of CCTV footage, often utilized for post-incident investigations. However, our proposed model offers a unique advantage: it can prevent crimes before they occur. By monitoring and analyzing real-time CCTV footage, the system triggers alerts to the relevant authorities if it detects any signs of a potential incident about to happen. By employing advanced technologies and machine learning algorithms, we can identify anomalous behavior, anticipate possible hazards, and adopt early actions. Criminal activity detection is constantly improving, which helps to promote security and create safer surroundings in addition to safeguarding assets.

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